

Lesson 1.8: Rational Functions and Zeros

Topic 1.8: Rational Functions and Zeros



Characteristics of Rational Functions

If $R(x) = \frac{N(x)}{D(x)}$ is a rational function and if $N(x)$ and $D(x)$ have **no common factors**, then the rational function R is said to be in **lowest terms** and we know these characteristics apply:

1. $R(x)$ has zeros when $N(x) = 0$, but not $D(x) = 0$.
2. $R(x)$ is undefined when $D(x) = 0$.

In the next lesson, we will discuss what happens if there is a common factor.

EX #1: Use the procedure above to find the zeros of the rational function. State the end-behavior using limit notation.

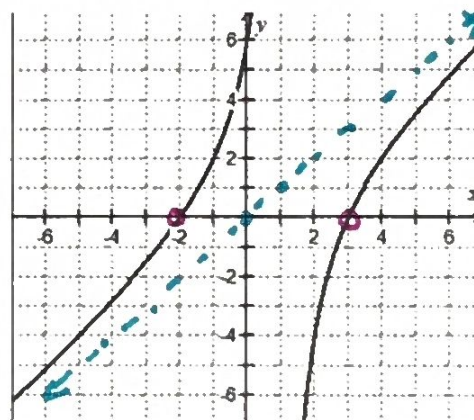
$$R(x) = \frac{x^2 - x - 6}{x - 1}$$

$$\begin{array}{r|rrrr} 1 & 1 & -1 & -6 & \\ & \downarrow & & & \\ & 1 & 0 & & \end{array}$$

SA: $y = x$

$$\begin{aligned} x^2 - x - 6 &= 0 \\ (x - 3)(x + 2) &= 0 \\ x &= 3, -2 \leftarrow \text{zeros} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} R(x) &= -\infty \\ \lim_{x \rightarrow \infty} R(x) &= \infty \end{aligned}$$



EX #2: Find the real zeros of the rational function, f . State the domain.

$$f(x) = \frac{x^3 - 3x^2 - 10x}{x^2 - x - 6}$$

Zeros: $x = 0, 5$

Domain: $(-\infty, -2) \cup (-2, 3) \cup (3, \infty)$

$$\begin{aligned} x(x^2 - 3x - 10) &= 0 \\ x(x - 5)(x + 2) &= 0 \\ x &= 0, 5, -2 \end{aligned}$$

$$\begin{aligned} x^2 - x - 6 &\neq 0 \\ (x - 3)(x + 2) &\neq 0 \\ x &\neq 3, -2 \end{aligned}$$

* Since $x \neq -2$, we cannot include it with the zeros.



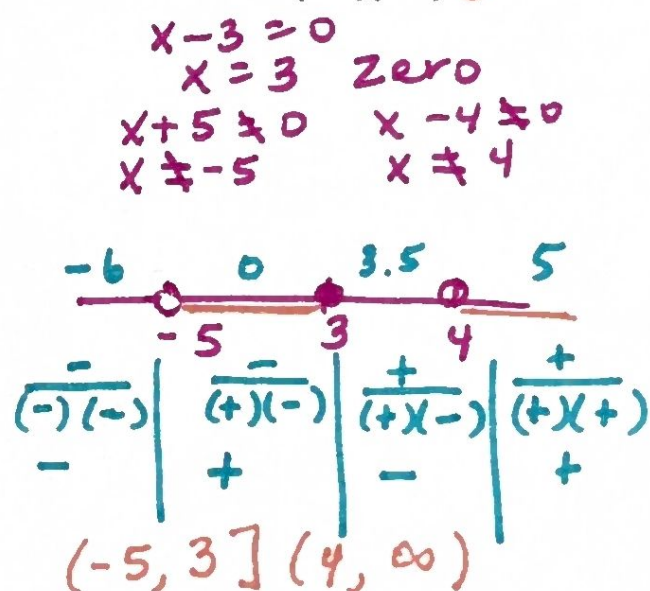
Solving Rational Inequalities

When solving rational inequalities, it is important to locate **BOTH** the **zeros** and the **undefined values**. We can use a procedural approach to solve rational inequalities.

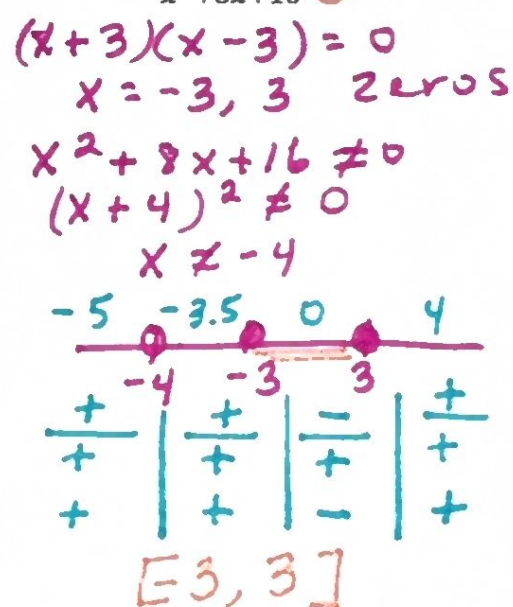
Remember: $f(x) > 0$ means the graph of f is above the x -axis and $f(x) < 0$ means the graph of f lies below the x -axis.

1. **Isolate** the inequality so that 0 is on the right.
2. Be sure the left side of the inequality is a single rational function $R(x) = \frac{N(x)}{D(x)}$.
3. **Factor** the expressions $N(x) = 0$ and $D(x) = 0$, use solutions to find values for the sign chart.
4. Create a **sign chart** with the values from **Step 3**. Mark the values where the denominator is zero. *These values will never be included in the solution.*
5. **Test values** in each interval to determine **positive or negative** values for the interval.
6. **Interpret** the sign chart results to answer the given inequality.
7. Write your answer in **interval notation**, consider the **endpoints**.

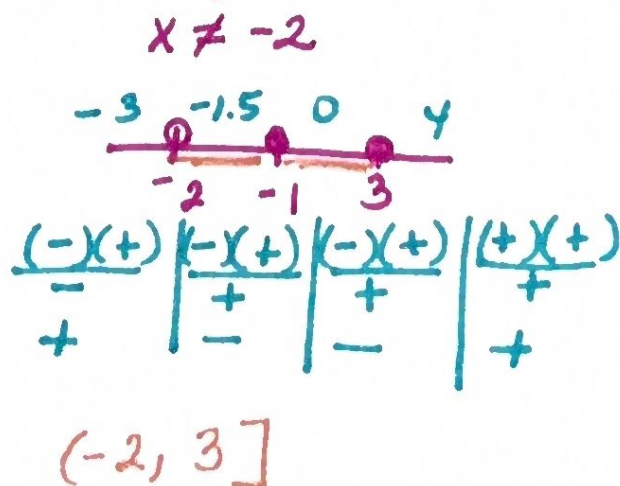
EX #3: Solve $\frac{x-3}{(x+5)(x-4)} \geq 0$



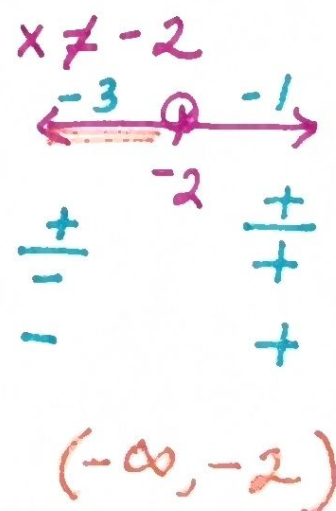
EX #4: Solve $\frac{x^2-9}{x^2+8x+16} \leq 0$

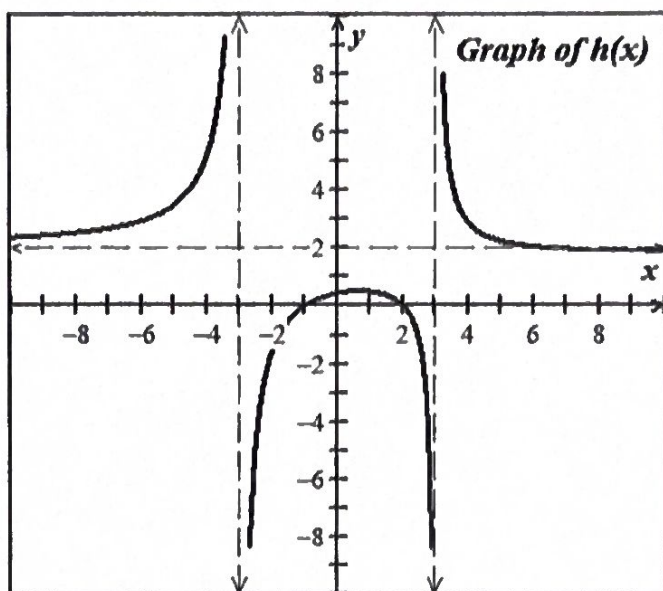


EX #5: Solve $\frac{(x-3)(x+1)^2}{x+2} \leq 0$ *use brackets for zeros only*



EX #6: Solve $\frac{5}{x+2} < 0$ *use () only*





EX #7: The graph of the rational function h is shown above. Given: $h(x) = \frac{2(x-2)(x+1)}{x^2-9}$.

Use the graph to solve the following inequalities.

A. $h(x) \leq 0$ *use brackets for zeros only*

$(-3, -1] [2, 3)$

$x = 2, -1$ zeros
 $x^2 - 9 \neq 0$
 $x \neq \pm 3$

B. $h(x) \geq 2$
 $(-\infty, -3) \cup (3, \infty)$

-4	-2	0	2.5	4
○	●	●	○	
-3	-1	2	3	
$\frac{(-)(-)}{+}$	$\frac{(-)(-)}{-}$	$\frac{(-)(+)}{-}$	$\frac{(+)(+)}{-}$	$\frac{(+)(+)}{+}$
+	-	+	-	+
✓ $\frac{2(-6)(-3)}{7}$	✗ $\frac{2(-2)(1)}{-9}$		✓ $\frac{2(2)(5)}{7}$	

C. $h(x) > 0$ *use () only*

$(-\infty, -3) \cup (-1, 2) \cup (3, \infty)$